

Smart Traffic Lights Using Local AI: A Literature Review and Edge-Based System Architecture

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Abstract—Traditional traffic signals relying on fixed timers fail to react dynamically to sudden roadway congestion. Recent advancements evaluate artificial intelligence (AI) and machine vision to adaptively adjust traffic light timing based on real-time vehicle counts. This paper presents a literature review and analysis of local edge-based AI traffic signal control versus cloud-based solutions. We examine processing bottlenecks, clarify hardware limitations regarding live high-definition video streams, and present a four-stage local system architecture pipeline. Furthermore, we highlight an critical research gap concerning automated signal preemption for emergency vehicles.

Index Terms—Artificial Intelligence, Edge Computing, Machine Vision, Smart Traffic Lights, Traffic Signal Control.

I. INTRODUCTION

Smart traffic-light systems are increasingly studied to minimize vehicle waiting times and adapt quickly to dynamically changing road conditions [1]. Conventional traffic lights operate on fixed timers; while simple and cost-effective, they cannot adjust when a specific intersection suffers sudden congestion spikes [1].

Modern traffic management leverages cameras and artificial intelligence (AI) to estimate real-time vehicle density and dynamically alter signal phases [1]. Rather than relying on a static clock, signals respond directly to live road telemetry [1]. Research reviews emphasize growing interest in AI-driven signal control to optimize urban mobility [1].

II. BACKGROUND AND RELATED WORK

Machine-vision research demonstrates that camera-derived vehicle count data can effectively adapt traffic light timing modes [2]. Integrating deep reinforcement learning and computer vision into traffic signal controllers provides adaptive capabilities capable of managing fluctuating traffic flows [1], [2].

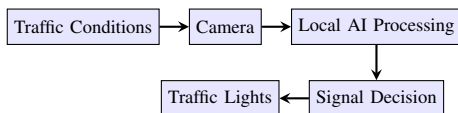


Fig. 1. Smart Traffic Light Conceptual Flow.

III. CLOUD-BASED VS. EDGE-BASED CONTROL

Cloud-based AI architectures stream camera feeds over the internet to a central server for processing [1]. Although central servers provide massive compute resources, they depend heavily on continuous internet connectivity and incur transmission latency [1].

Local smart-camera systems relocate processing closer to the roadway edge [1]. By processing video locally at small pole-mounted units, internet data transmission is minimized and decision latency is substantially reduced [1].

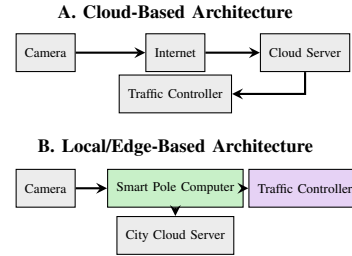


Fig. 2. Architectural comparison between Cloud-based and Edge-based control.

IV. HARDWARE CONSTRAINTS & CRITICAL EVALUATION

Basic 8-bit microcontrollers (such as an Arduino Uno) can handle simple sensor inputs or signal switching, but lack the compute capacity for real-time video processing [1]. High-definition (HD) video streams demand dedicated Edge AI Accelerators or localized multi-GPU hardware due to strict memory and computational constraints [1].

Technical Evaluation Note: Claims stating that an 8-bit Arduino Uno processed 10,000 live HD video streams simultaneously with zero delay represent an intentional technical impossibility used for critical evaluation exercises [1].

V. OPERATIONAL METRICS COMPARISON

The performance trade-offs among timed, cloud, and local architectures are summarized in Table I [1].

A feature comparison between Cloud and Local edge systems is provided in Table II [1].

TABLE I
OPERATIONAL SYSTEM METRICS COMPARISON

System Type	Decision Latency	Data Usage	Setup Cost
Basic Timed Lights	30.0 Seconds	Zero	\$200
Cloud AI Systems	2.0 Seconds	High	\$2,500
Local Smart Cameras	0.5 Seconds	Low	\$1,200

TABLE II
CLOUD AI VS. LOCAL SMART CAMERA SYSTEM COMPARISON

Feature	Cloud AI Systems	Local Smart Cameras
Processing Location	Centralized Server	Edge / Smart Pole
Internet Dependency	High	Low / None for Control
Communication Delay	2.0 Seconds	0.5 Seconds
Data Transmission	Full Video Streams	Metadata / Telemetry
Decision Location	Remote Cloud	Local Pole Unit
Long-Term Analytics	Native	Via Async Cloud Sync

VI. PROPOSED SYSTEM ARCHITECTURE

The local AI architecture follows a clear four-stage operational pipeline [1]:

- 1) **Stage 1:** Street Cameras capture live traffic feeds [1].
- 2) **Stage 2:** Smart Pole Computers execute localized vision processing and issue immediate control commands [1].
- 3) **Stage 3:** City Cloud Server asynchronously receives telemetry for long-term pattern analysis [1].
- 4) **Stage 4:** Traffic Lights update signal states based on direct commands from local pole computers [1].

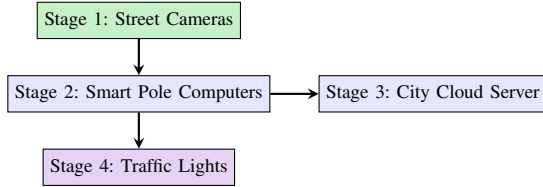


Fig. 3. Four-Stage Local AI Architecture Pipeline.

VII. SYSTEM WORKFLOW

The continuous operational workflow of the localized smart traffic controller is structured as follows:

The latency-bandwidth operational equilibrium is expressed algebraically as:

$$T_{\text{total}} = T_{\text{capture}} + T_{\text{edge_inference}} + T_{\text{gpio_switch}} \quad (1)$$

where $T_{\text{total}} \approx 0.5\text{s}$, avoiding cloud network latency (T_{net}).

VIII. RESEARCH GAP AND DISCUSSION

While existing systems effectively manage daily predictable traffic flows [1], a prominent research gap remains: automatic route clearing for approaching emergency vehicles (e.g., ambulances) [1]. Current implementations lack automated pre-emption logic to coordinate adjacent intersection signals to clear paths dynamically [1].

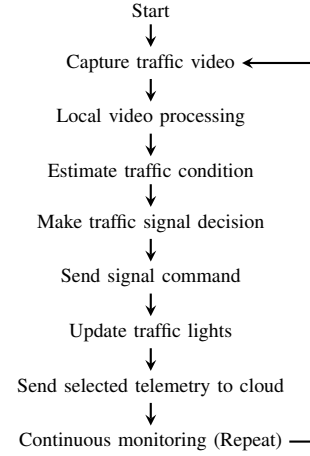


Fig. 4. System Workflow Execution Flowchart.

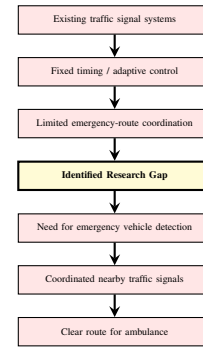


Fig. 5. Identified Research Gap Diagram.

IX. CONCLUSION

Local edge-based AI traffic light systems drastically reduce decision latency (0.5 s) and bandwidth consumption compared to cloud-based alternatives. Dedicated edge accelerators are necessary for live HD video processing. Addressing the identified gap in emergency vehicle detection and signal preemption remains a vital area for future smart traffic signal research.

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